Mahdee S. Khan

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Education

Stony Brook University — Stony Brook, New York
B.S. in Computer Engineering, GPA: 3.43/4.00 — *Expected 2029*

- **VLSI & Microarchitecture:** VLSI System Design, Digital Design Using VHDL and PLDs, Fundamental Algorithms for Automated Electronic Design and Machine Learning Hardware
- **Computer Architecture & Systems:** Digital Systems Architecture, Computer Systems Architecture, Digital System Design

MDQ Academy — Commack, New York
High School, GPA: 5.02/5.00 — *Sep 2021 – Jun 2025*

- Graduated with a Regents Diploma with Advanced Designation
- Valedictorian

Experience

MACH1 Innovation Hub — Stony Brook, NY
Data Analyst, Business Management Team — *Jan 2026 – Present*

- Designed a user-focused survey to collect feedback from individuals who use The Space; determined which types of data would be most useful for future analysis and service improvements and created survey questions focused on user demographics, including major, experience level, and interests.
- Collected input on what users wanted to see added, changed, or improved within The Space and helped build a feedback tool intended to guide future decisions and improve the overall user experience. Used Microsoft Excel to clean, organize, and format survey data for future analysis.

Projects

RISC-V CPU Simulator with Verification Testbench — *Personal Project*
C, RISC-V RV32I ISA

- Designing a cycle-accurate CPU simulator in C for a subset of the RISC-V RV32I instruction set, implementing the fetch, decode, and execute stages with simulated registers, memory, and a program counter.
- Building an automated testbench of C-based checker functions to validate CPU register and memory outputs against expected results across arithmetic, branching, and memory-access test cases, mirroring verification methodologies used in industry hardware verification workflows.

Formal Verification of a Synchronous FIFO — *Personal Project*
SystemVerilog (SVA), SymbiYosys

- Designed and simulated a parameterized synchronous FIFO in Verilog, then wrote SystemVerilog Assertions (SVA) specifying functional correctness properties including no overflow, no underflow, and preservation of data ordering.
- Used SymbiYosys to run bounded model checking (BMC) and k-induction proofs verifying the FIFO satisfies all specified properties across every reachable state, and added cover properties to confirm the assertions were not vacuously true.

Technical Skills

Programming: Verilog, System Verilog, C++, C, Python, LT Spice, Matlab, Java, MySQL
Technical: FPGA (Xilinx Zynq), Xilinx Vivado, Catapult HLS, UVM, TCL, Cadence JasperGold, SymbiYosys, Vivado, mflowgen, Synopsys Design Compiler, LTspice, AWS (EC2, S3, RDS, API Gateway), Git